Moving Averages

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Introduction

A moving average smooths a time series by replacing each observation with an average of nearby values. It is the oldest and simplest tool for highlighting trend by suppressing high-frequency noise; it underpins more sophisticated techniques (X-11 decomposition, Henderson trend filters) and is fundamental to financial chart analysis. The choice between trailing, centred, and weighted moving averages governs the trade-off between smoothness and lag, and choosing the right window size for the application is the principal craft of moving-average smoothing.

Prerequisites

A working understanding of time series, the trade-off between smoothness and lag, and the role of window size in noise suppression.

Theory

Three main variants:

- **Simple moving average**: arithmetic mean of k consecutive observations. Trailing alignment uses the last k values up to time t (one-sided, introduces lag); centred alignment averages $(k-1)/2$ before and after (two-sided, no lag but undefined at the boundaries).
- **Weighted moving average**: assigns weights to each observation in the window — triangular, Hamming, Spencer 15-term (used in classical seasonal adjustment), or any user-defined kernel.
- **Exponential moving average**: geometrically declining weights, $\hat{y}_t = \alpha y_t + (1-\alpha)\hat{y}_{t-1}$. Closely related to simple exponential smoothing.

A centred MA with window equal to the seasonal period eliminates the seasonal component, leaving trend plus residual; this is the foundation of classical seasonal decomposition.

Assumptions

Regular sampling intervals (irregular spacing requires alternative methods); stationarity is not required, but the moving-average filter assumes a slowly-varying trend.

R Implementation

```r
library(zoo); library(TTR)

x <- AirPassengers

# Trailing mean
sm_3 <- rollmean(x, 3, align = "right", fill = NA)

# Centred 12-month MA (removes annual seasonality)
sm_12 <- rollmean(x, 12, align = "center", fill = NA)
```

```r
# Weighted MA
sm_w <- TTR::WMA(x, n = 12, wts = 1:12)

plot(x); lines(sm_12, col = "red", lwd = 2)
```

Output & Results

`rollmean()` returns the smoothed series with NA at boundaries (trailing alignment leaves NAs at the start; centred alignment leaves NAs at both ends). `TTR::WMA()` produces weighted alternatives. Plotting the centred 12-month MA on top of the raw series reveals the trend that the seasonal cycle hides.

Interpretation

A reporting sentence: "A centred 12-month moving average revealed the secular trend in monthly air-passenger counts, with the smoothed series showing approximately 5 % annual growth and the seasonal cycle largely eliminated by averaging over a full period." Quote the window size and alignment in any report.

Practical Tips

- A centred moving average of window equal to the seasonal period eliminates the seasonal cycle exactly; this is the X-11 first step.
- Trailing moving averages introduce lag proportional to half the window size; use only when real-time estimation is needed and the lag is acceptable.
- For smoother curves with minimal lag, exponential smoothing, LOESS, or spline smoothers are often preferable to wide moving averages.
- Centred MA is undefined at the start and end of the series; for the boundary, use one-sided MA, asymmetric weights, or an end-effects correction.
- Weighted MAs with carefully chosen kernels (Spencer's 15-term, Henderson's filters) are used in official-statistics seasonal adjustment because they preserve cubic polynomial trends while suppressing irregular components.
- Pair MA smoothing with the original series on the same plot; the gap between them is the seasonal-plus-irregular component.

R Packages Used

`zoo::rollmean()` and `rollmeanr()` for simple moving averages with control over alignment; `TTR::SMA()`, `WMA()`, and `EMA()` for technical-analysis moving averages including weighted and exponential; `forecast::ma()` for centred MAs with seasonal-period defaults; `slider::slide_dbl()` for tidyverse-friendly sliding-window computations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Time series basics